Nucleotides Metabolism

Nucleotide Biosynthesis

- Nucleotides are precursors of DNA and RNA
- Are essential carriers of chemical energy: ATP/GTP
- Are components of cofactors: NAD^+, FAD, SAM, CoA
- Are activators of biosynthetic intermediates: UDP-Glu

  CDP-DAG
- Some are also cellular second messengers: cAMP, cGMP

Nucleotide Biosynthesis

- Nucleotides can be synthesized de novo ("from the beginning") from amino acids, ribose-5-phosphate, CO_2, and NH_3.
- Nucleotides can be salvaged from RNA, DNA, and cofactor degradation.
- Many parasites (e.g., malaria) lack de novo biosynthesis pathways and rely exclusively on salvage.
  - Compounds that inhibit salvage pathways are promising antiparasite drugs.

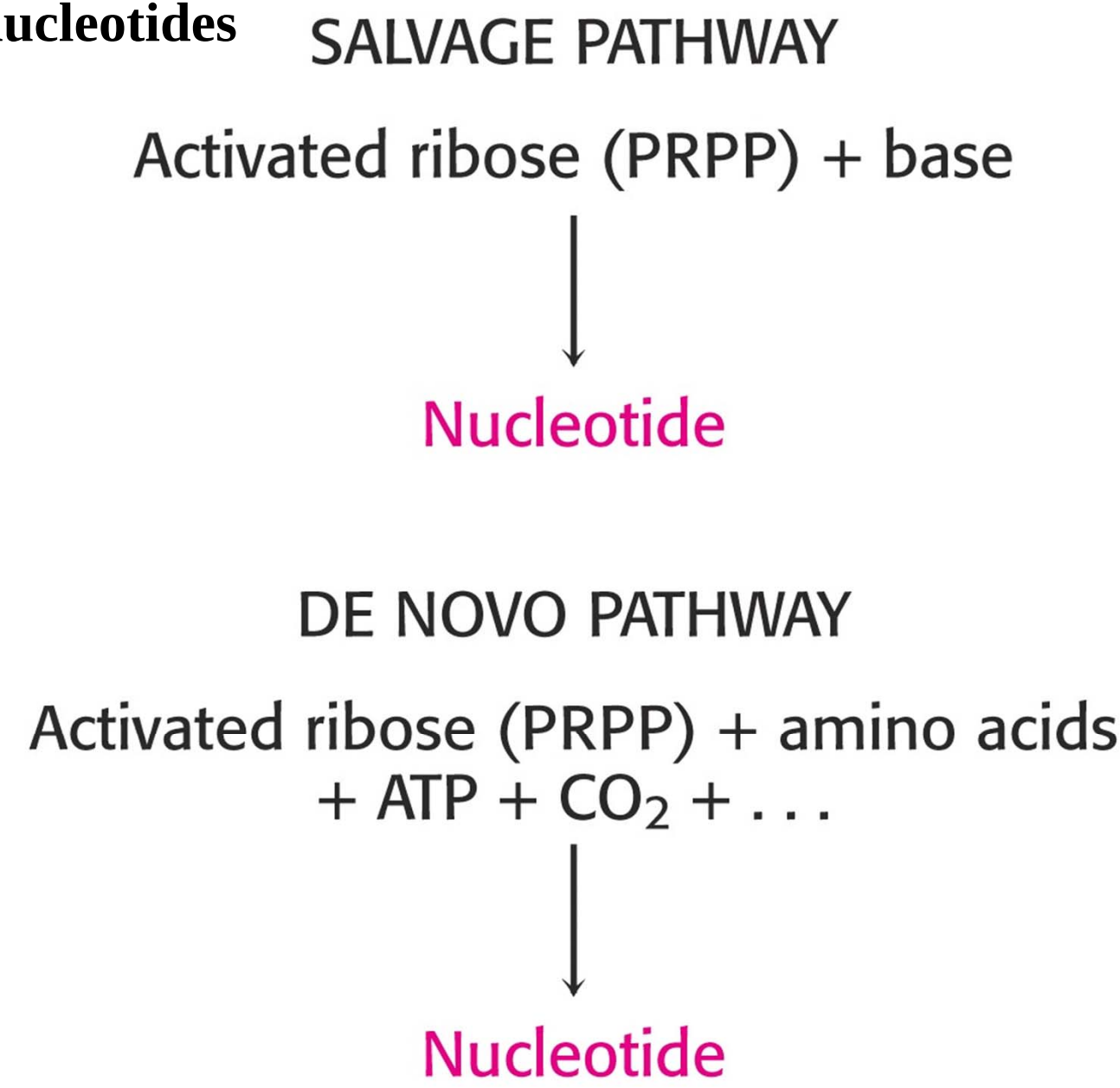
Many Chemotherapeutic Agents Target Nucleotide Biosynthesis

- Glutamine analogs: azaserine, acivicin (highly toxic)
  - inhibit glutamine amidotransferases
- Fluorouracil
  - converted by salvage pathway into FdUMP, which inhibits thymidylate synthase
- Methotrexate and aminopterin
  - inhibit dihydrofolate reductase (competitive inhibitors)

Antibiotics Also Target Nucleotide Biosynthesis

- Allopurinol, and so on
  - studied against African sleeping sickness (*trypanosomiasis*) because the trypanosomes lack enzymes for de novo nucleotide synthesis
- Trimethoprim
  - inhibits bacterial dihydrofolate reductase but binds human enzyme several orders of magnitude less strongly

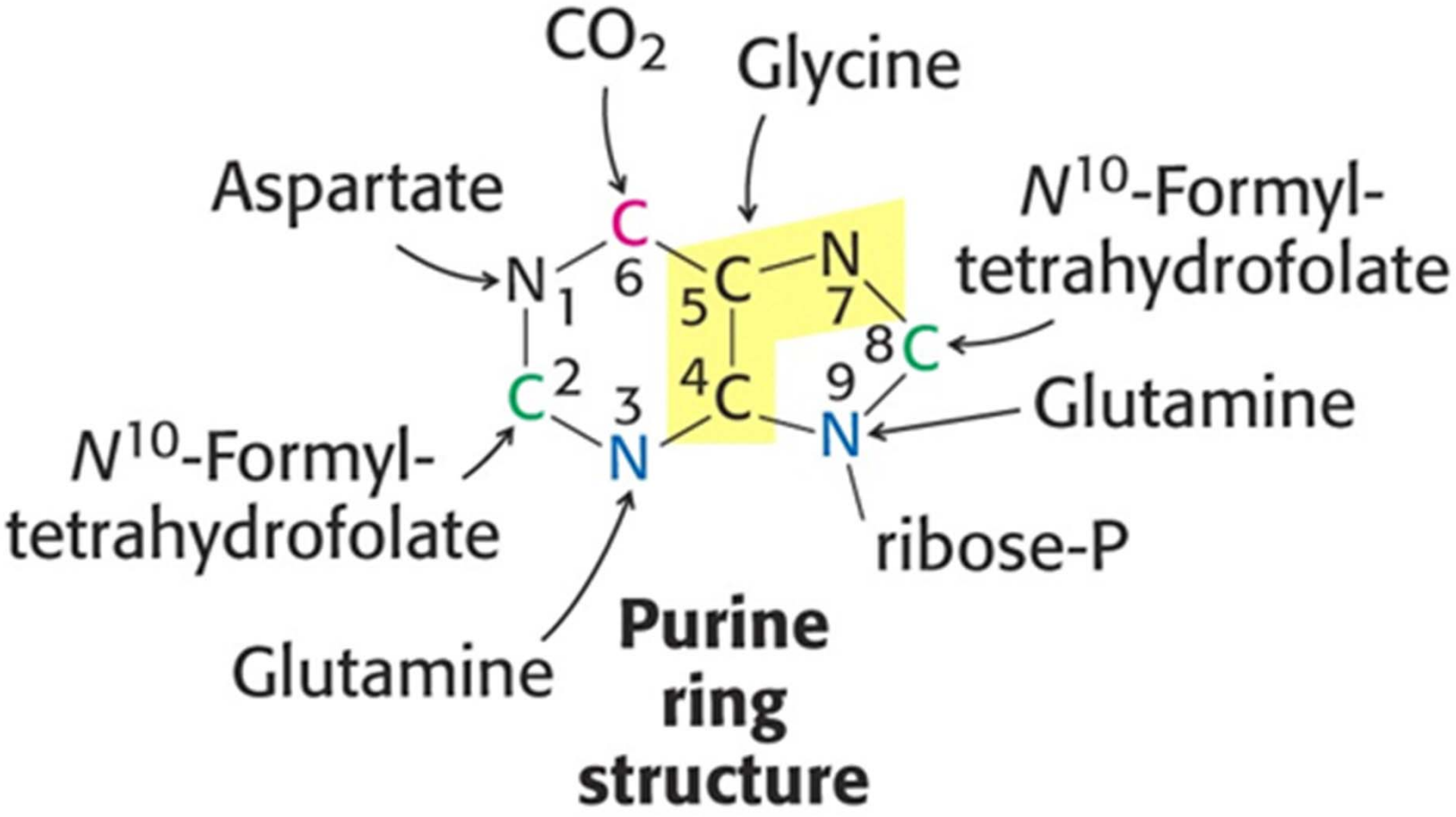
Metabolic pathways leading to formation of nucleotides



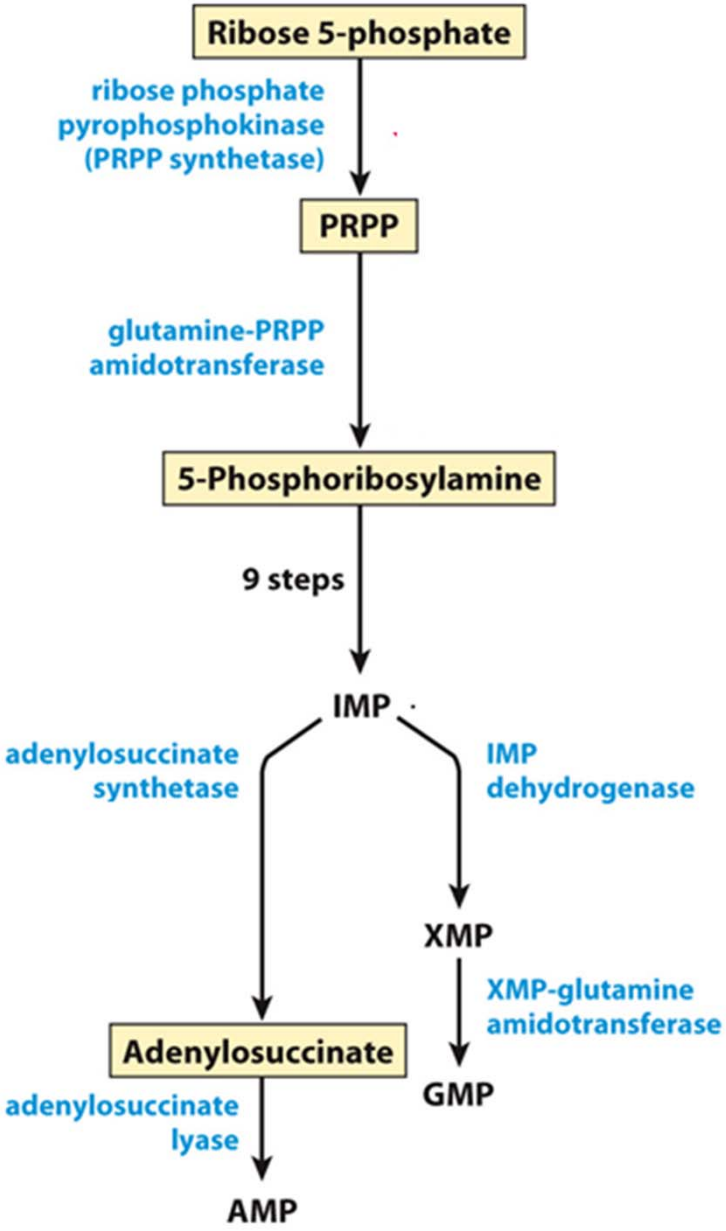
De Novo Biosynthesis of Nucleotides

- Approximately the same in all organisms studied
- Ribose-5 phosphate is a common precursor
- Gln and Asp provides most amino groups.
- Gly is precursor for purines
- High ATP consumming
- Nucleotide pools are kept low, so cells must continually synthesize them.
  - This synthesis may actually limit rates of transcription and replication.

Origin of Ring Atoms in Purines

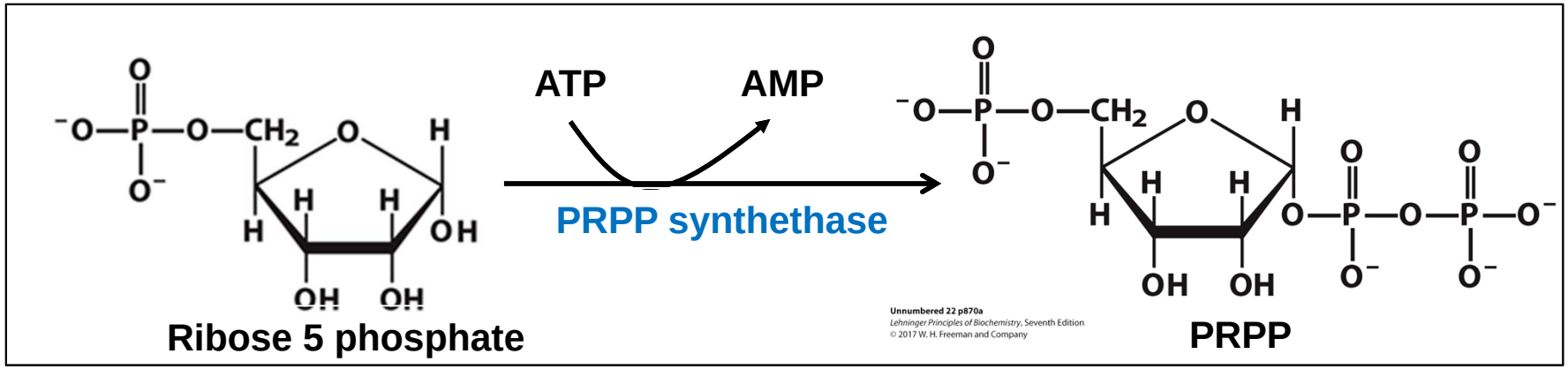


Overview of Purine nucleotides synthesis pathway



De Novo Biosynthesis of Purines Begins with PRPP

- Adenine and guanine are synthesized as AMP and GMP.
- Synthesis begins with reaction of 5-phosphoribosyl 1-pyrophosphate (PRPP) with Glu.
- Purine ring builds up following the addition of three carbons from glycine.
- The first intermediate with a full purine ring is **inosinate (IMP).**



De Novo Biosynthesis of Purines

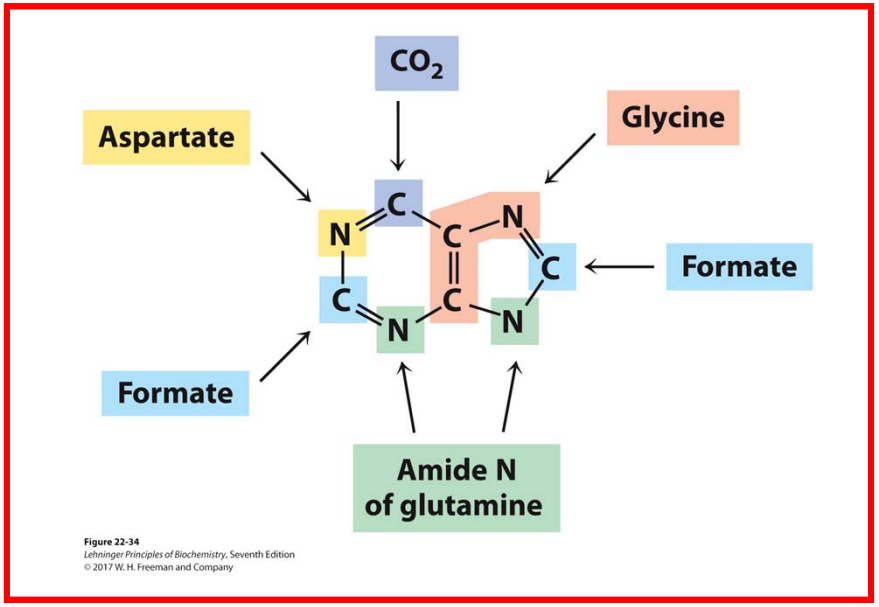
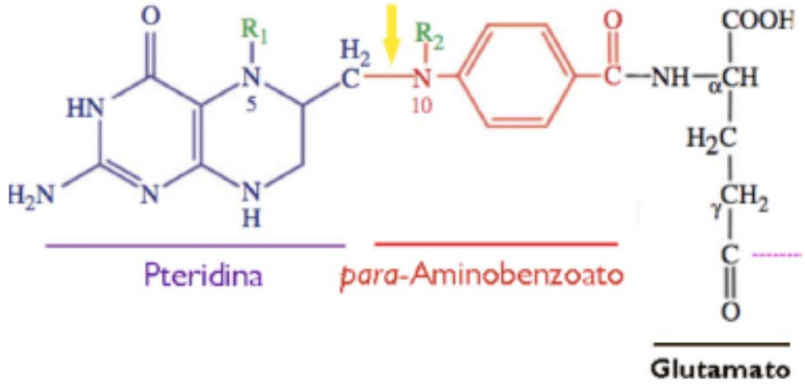


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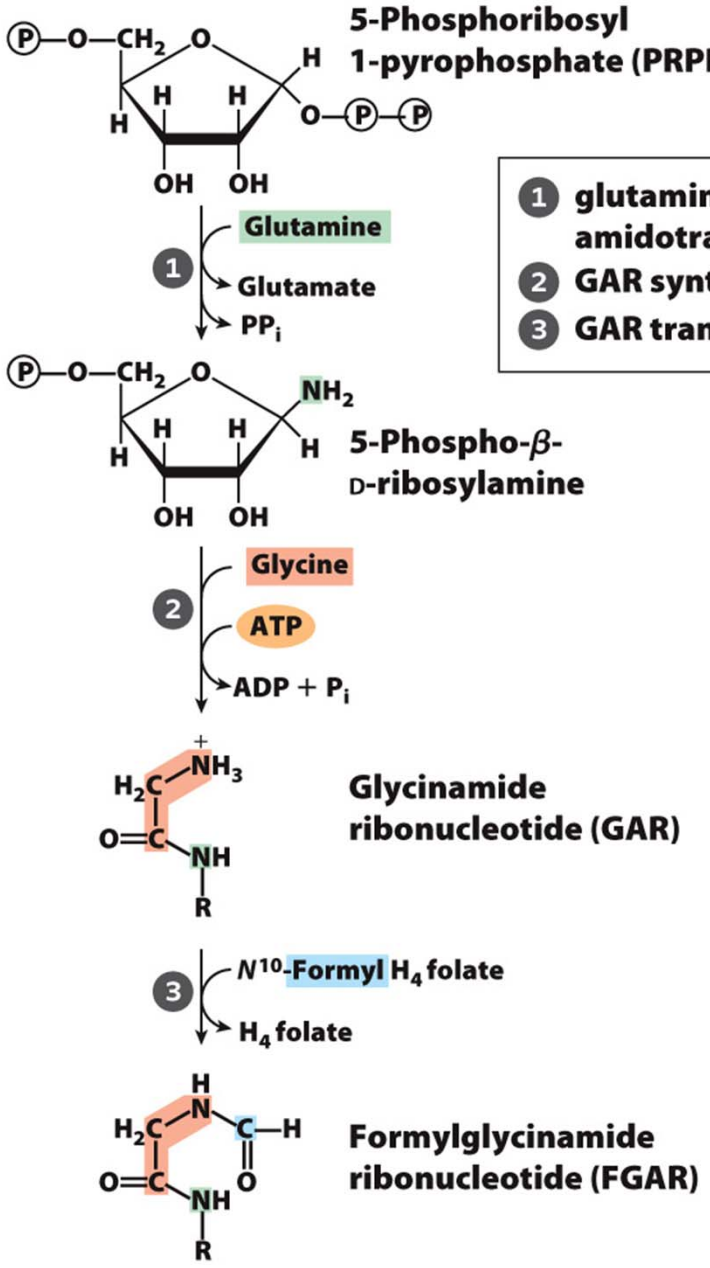


Figure 22-35 part 1
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Synthesis of AMP and GMP from IMP

Note that ATP is used to phosphorylate GMP precursor, while GTP is used to phosphorylate AMP precursor.

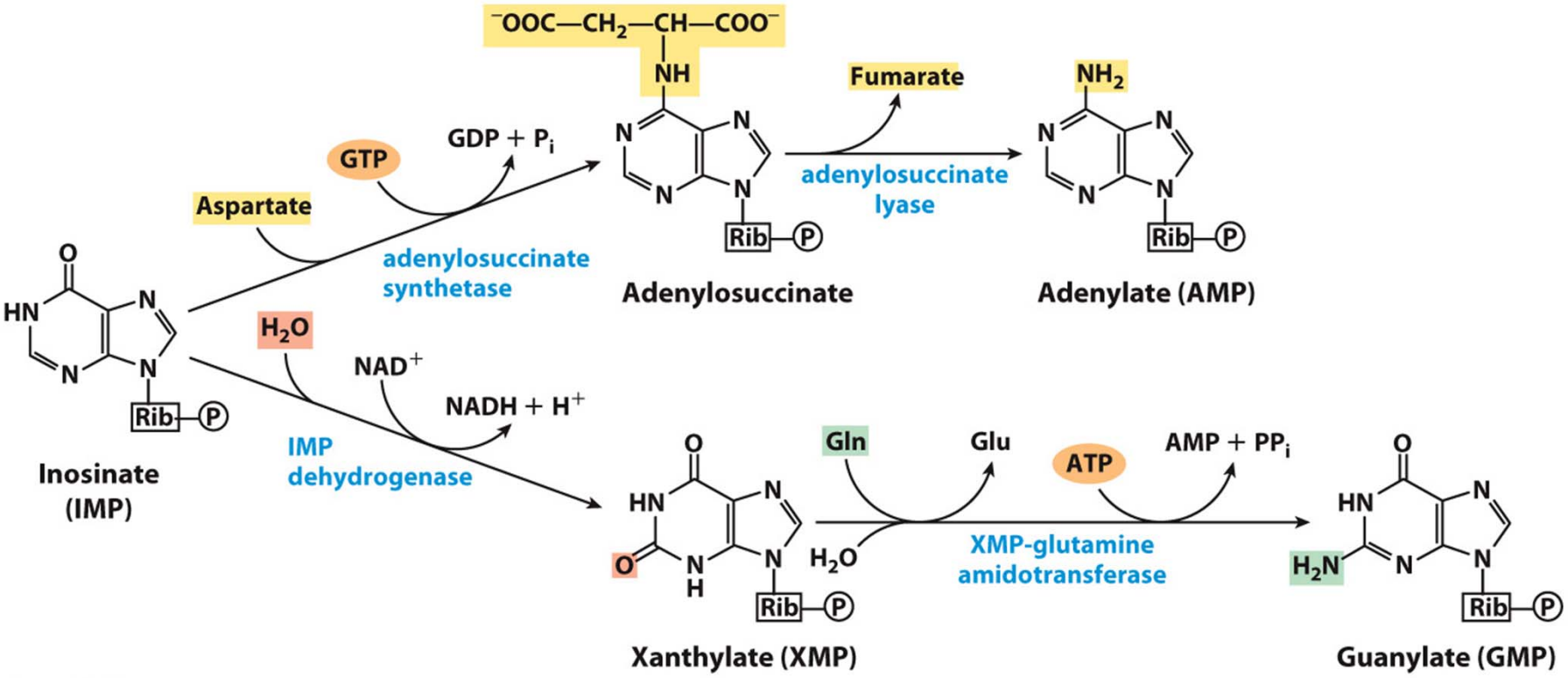


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Regulation of Purine Biosynthesis in *E. coli* Largely Consists of Feedback Inhibition

Four Major Mechanisms

1. *Glutamine-PRPP amidotransferase* is inhibited by end-products IMP, AMP, and GMP.
2. Excess GMP inhibits formation of xanthylate from inosinate by *IMP dehydrogenase.*
3. GMP and AMP concentrations inhibit phosphorylation steps.
4. PRPP synthesis is inhibited by ADP and GDP.

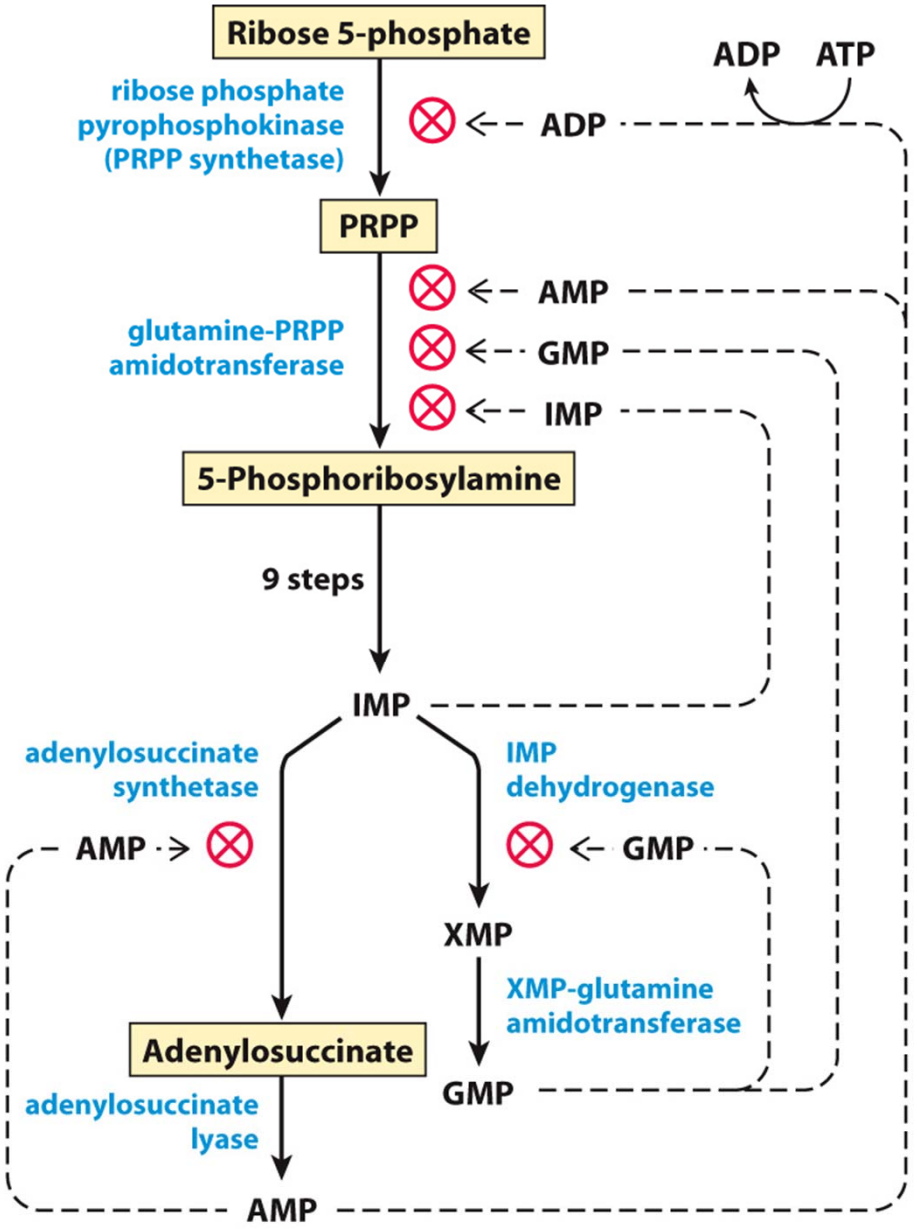
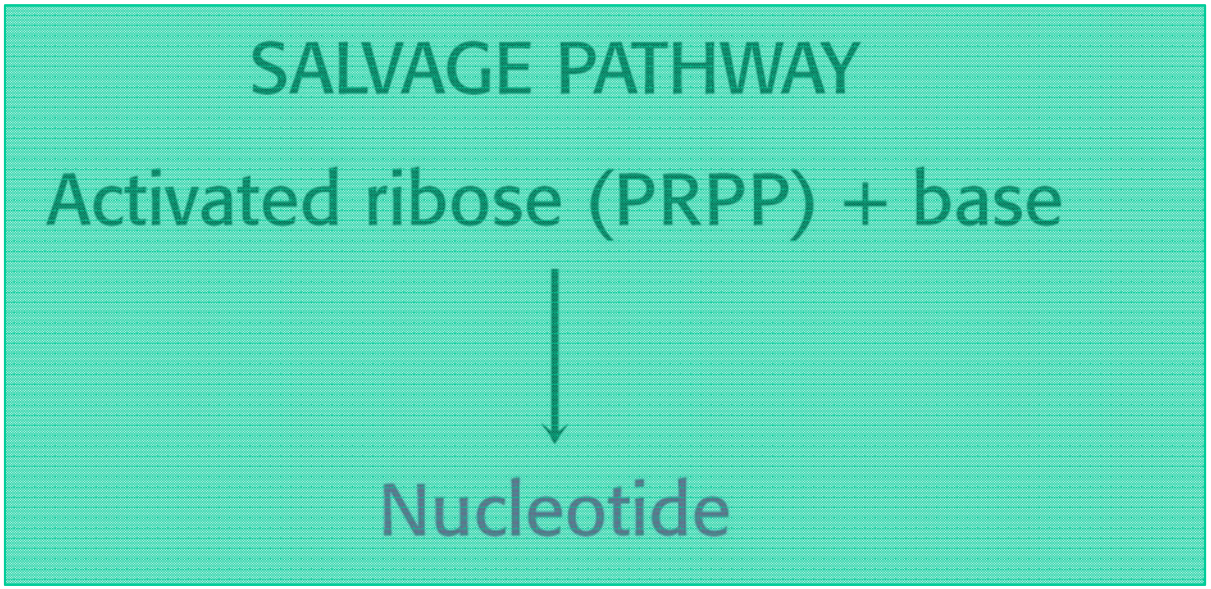


Figure 22-37
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Metabolic pathways leading to formation of nucleotides



DE NOVO PATHWAY

Activated ribose (PRPP) + amino acids + ATP + CO_2 + . . .

Nucleotide

Purine recycling pathways

- Free purines comming from diet or nucleotide catabolism can be reused to synthesize new purin nucleotides
- Purine recycling avoids the excessive uric acid (the final product of purines catabolism)

Purine recycling pathways

- Adenine phosphoribosyl transferase (APRT)

Adenine + PRPP→ AMP + PPi

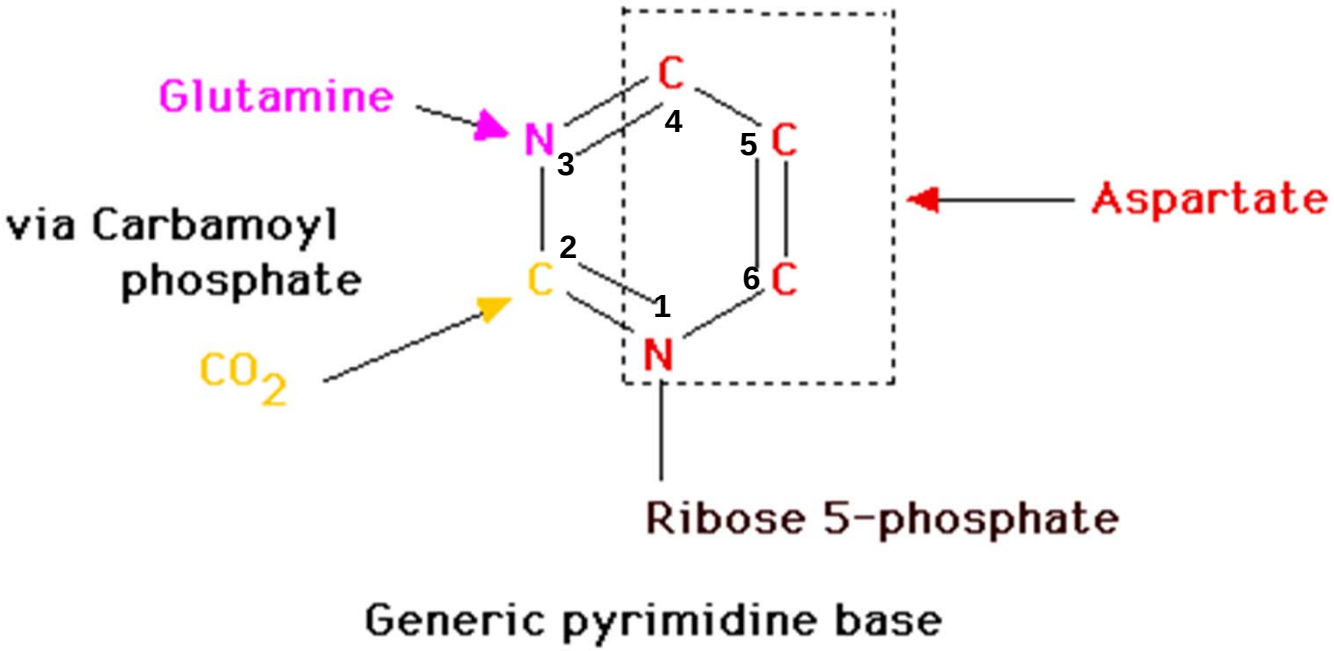
- Hipoxantine-Guanine phosphoribosyl transferase (HGPRT)

–Hipoxantine + PRPP→ IMP + PPi

–Guanine + PRPP→ GMP + PPi

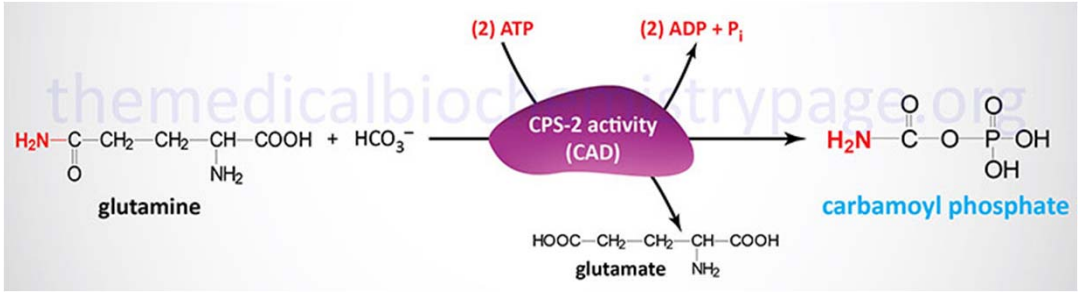
Lack of HGPRT activity causes a severe syndrome named Lesch_Nyham

De Novo Synthesis of Pyrimidine Nucleotides



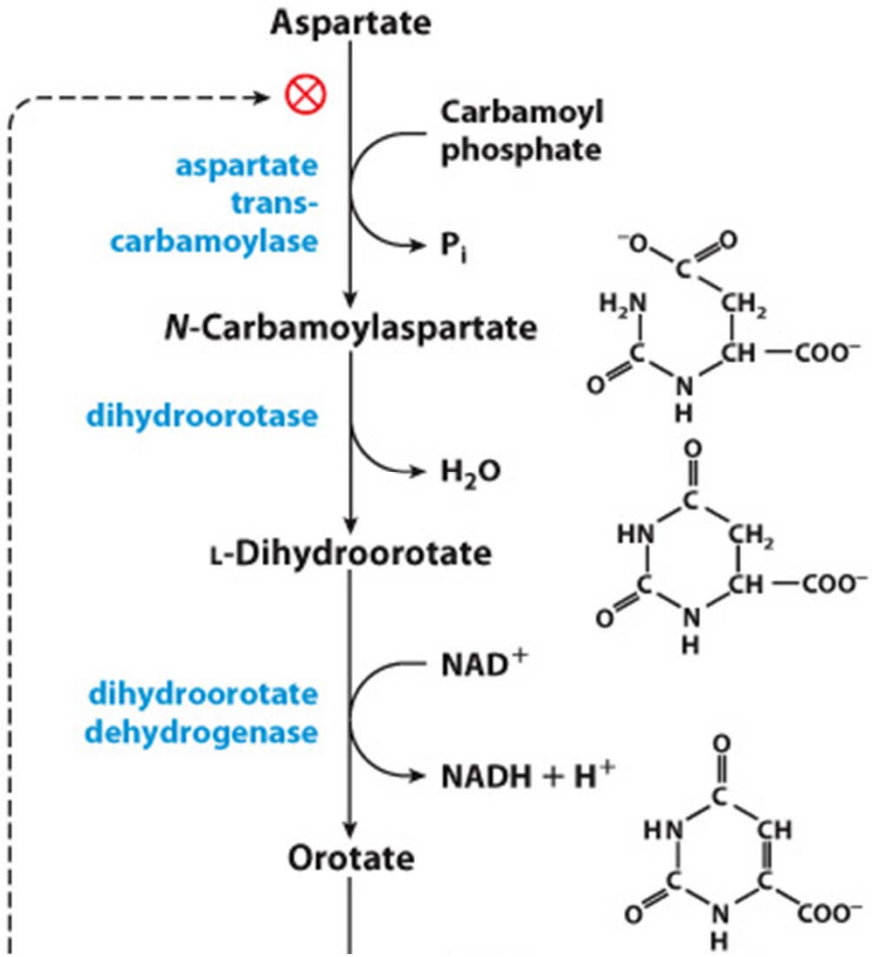
Generic pyrimidine base

De Novo Synthesis of Pyrimidine Nucleotides (1)



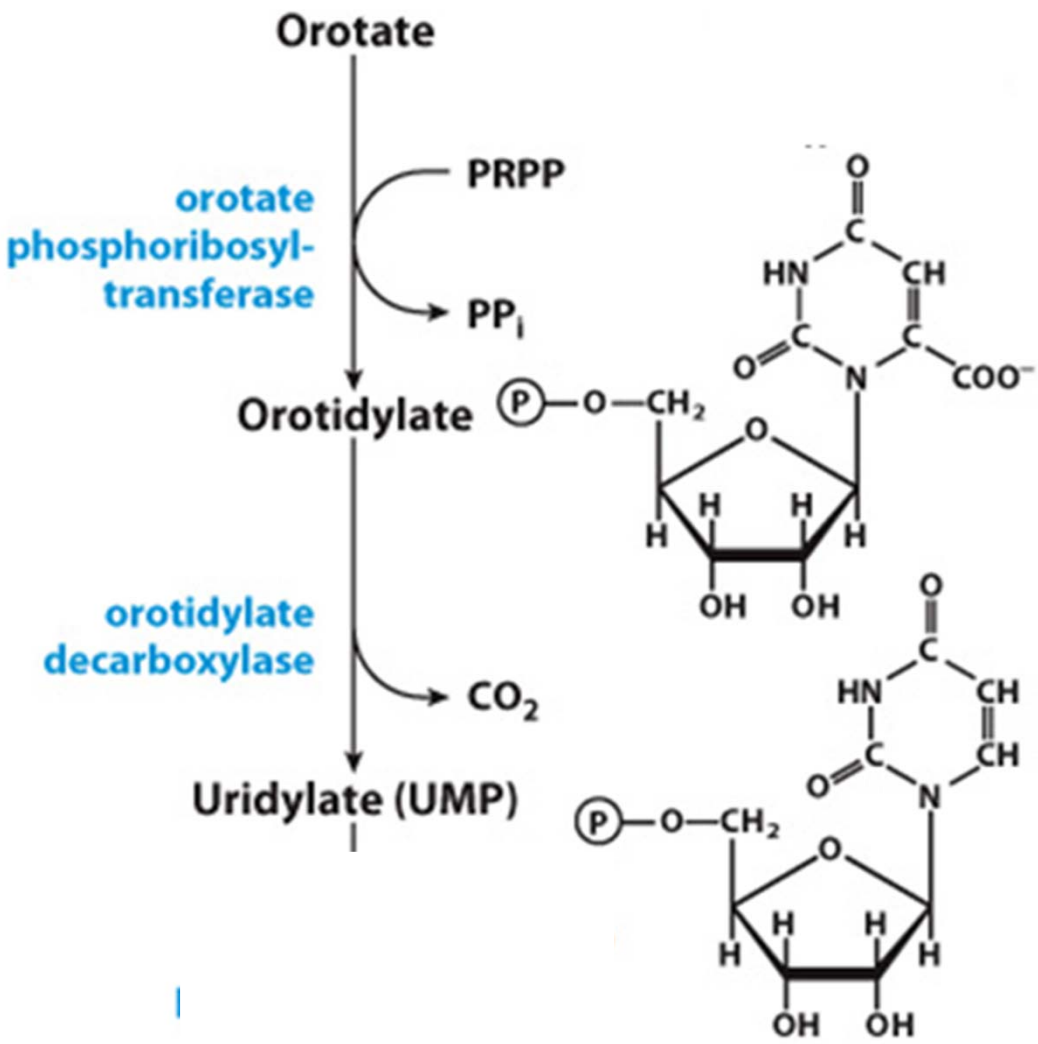
Unlike purine synthesis, pyrimidine synthesis proceeds *by first making the pyrimidine ring* (in the form of orotate) and *then* attaching it to ribose 5-phosphate.

Aspartate and carbamoyl phosphate provide the atoms for the ring structure.



De Novo Synthesis of Pyrimidine Nucleotides (2)

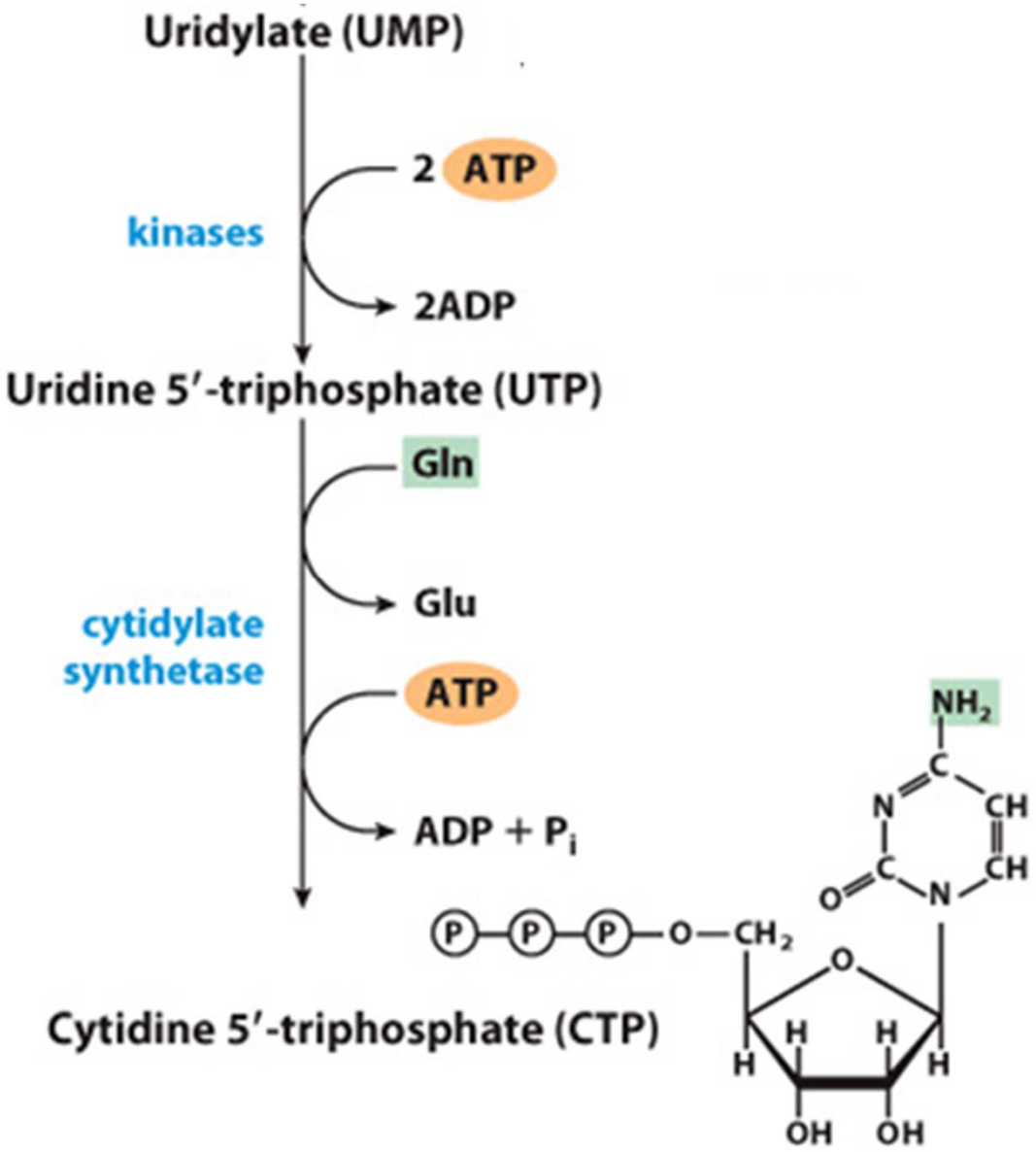
After addition of ribose-5-phosphate via PRPP, the resulting nucleotide (orotidylate) is decarboxylated to form uridylate (UMP), the first possible pyrimidine.



De Novo Synthesis of Pyrimidine Nucleotides (3)

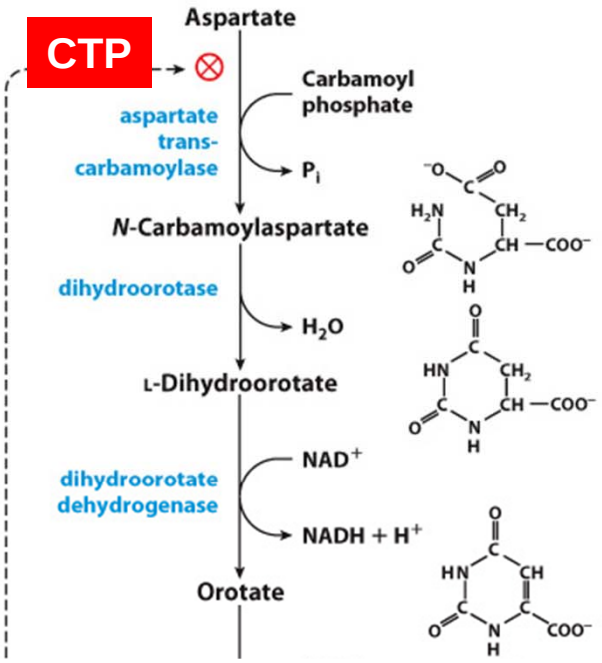
UMP is phosphorylated to UTP.

After formation of UTP, amination can convert UTP to CTP.



Regulation of Pyrimidine Biosynthesis Is Also via Feedback Inhibition

- ATCase is inhibited by end-product CTP and is accelerated by ATP.



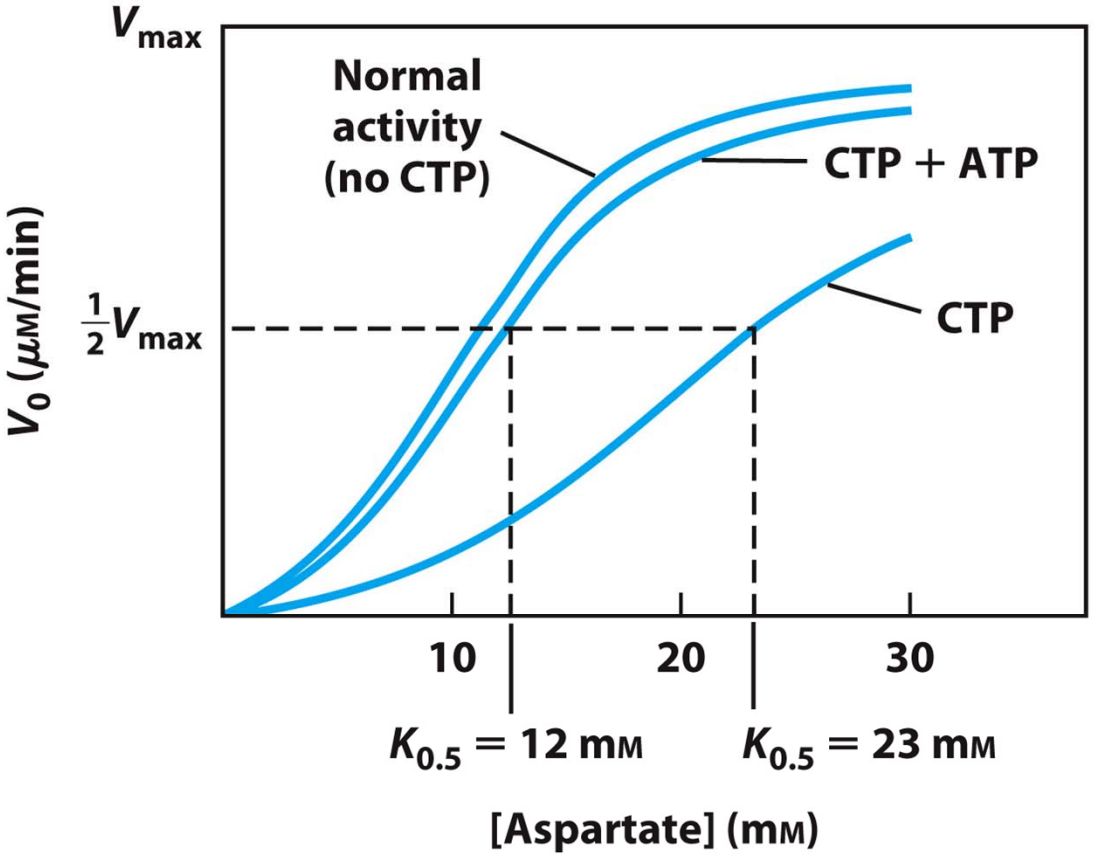


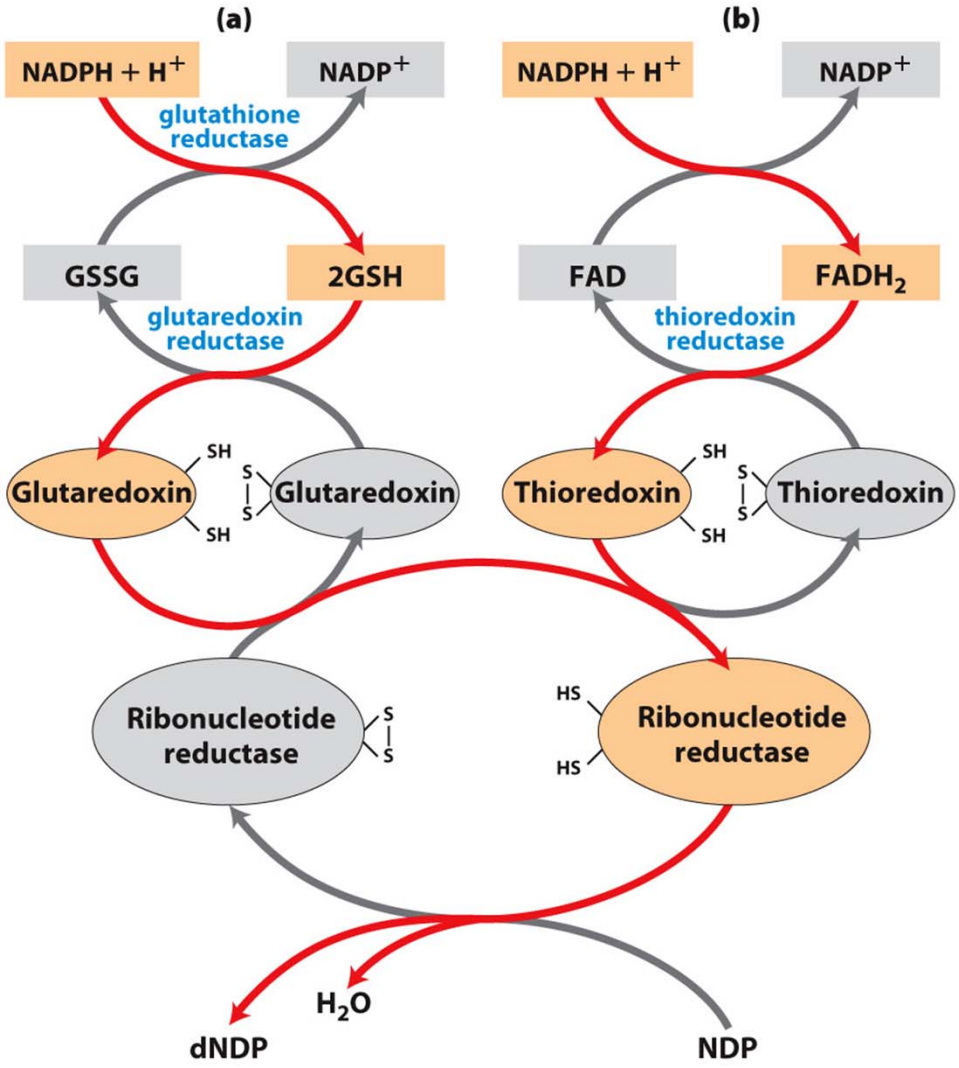
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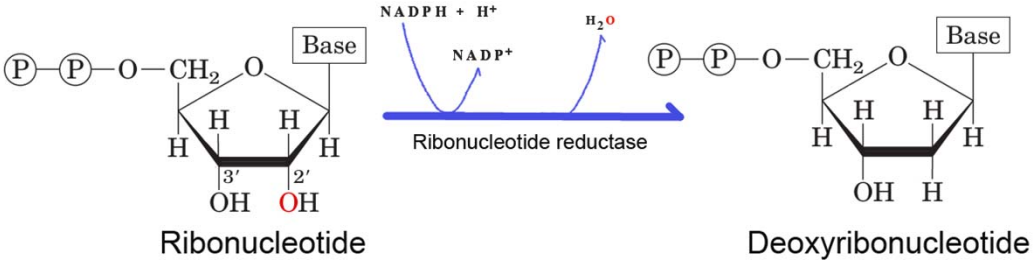
Ribonucleotides Are Precursors to Deoxyribonucleotides

- 2'C-OH bond is directly reduced to 2'-H bond… without activating the carbon!
  - catalyzed by *ribonucleotide reductase*
- Mechanism: Two H atoms are donated by NADPH and carried by proteins thioredoxin or glutaredoxin.

Reduction of Ribonucleotides by Ribonucleotide Reductase

- NADPH serves as the electron donor.
- Funneled through glutathione (a) or thioredoxin pathways (b)





Ribonucleotide Reductase Has Two Types of Regulatory Sites

- One type affects *activity.*
  - ATP activates, dATP inhibits
- The other type affects *substrate specificity* in order to maintain balanced pools of nucleotides.
  - If ATP or dATP high → less specificity for adenine and MORE specificity for UDP and CDP, and so on.
  - Enzyme ***oligomerizes*** to accomplish this change.

dTMP Is Made from dUTP

- Roundabout pathway…

1. dUTP is made.
2. dUTP → to dUMP by dUTPase
3. dUMP → dTMP by *thymidylate synthase*
   - adds a methyl group from *tetrahydrofolate*

Thymidylate synthase is a target for some anticancer drugs.

Conversion of dUMP to dTMP by Thymidylate Synthase

Thymidylate synthesis and folate metabolism are targets of chemotherapy

-**Fluorouracil** is activated by cells into **FdUMP** which inhibits **thymidilate synthase**

-**Methotrexate, Aminopterin** (chemotherapy) and **Trimethoprim** (antibiotic) are analogs of dihydrofolate and inhibit **dihydrofolate reductase**

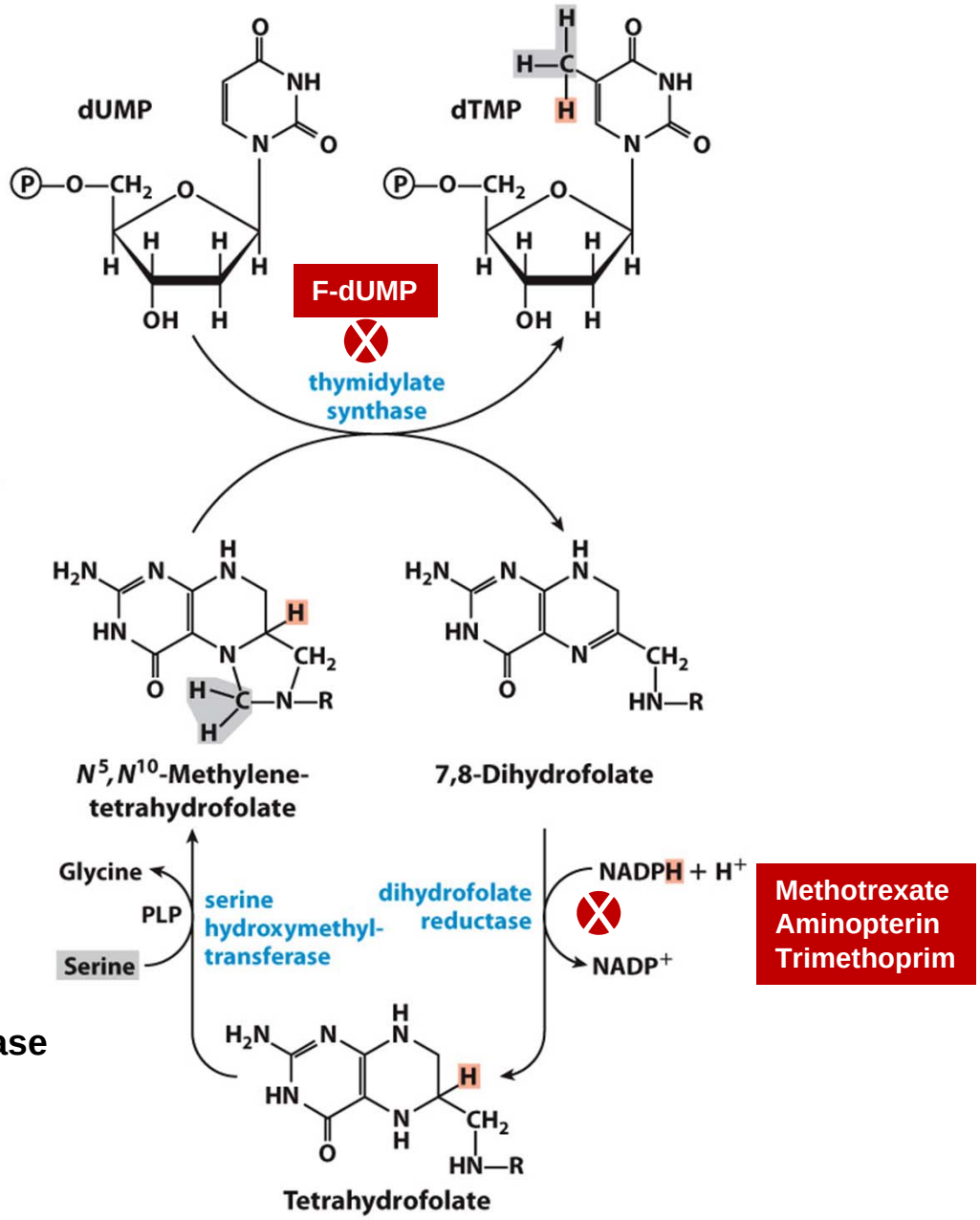


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Folic Acid Deficiency Leads to Reduced Thymidylate Synthesis

- Folic acid deficiency is widespread, especially in nutritionally poor populations.
- Reduced thymidylate synthesis causes uracil to be incorporated into DNA.
- Repair mechanisms remove the uracil by creating strand breaks that affect the structure and function of DNA.
  - associated with cancer, heart disease, neurological impairment

Catabolism of Purines

1. Dephosphorylation (via *5'-nucleotidase*)
2. Deamination and hydrolysis of ribose lead to production of xanthine.
3. Hypoxanthine and xanthine are then oxidized into uric acid by ***xanthine oxidase.***

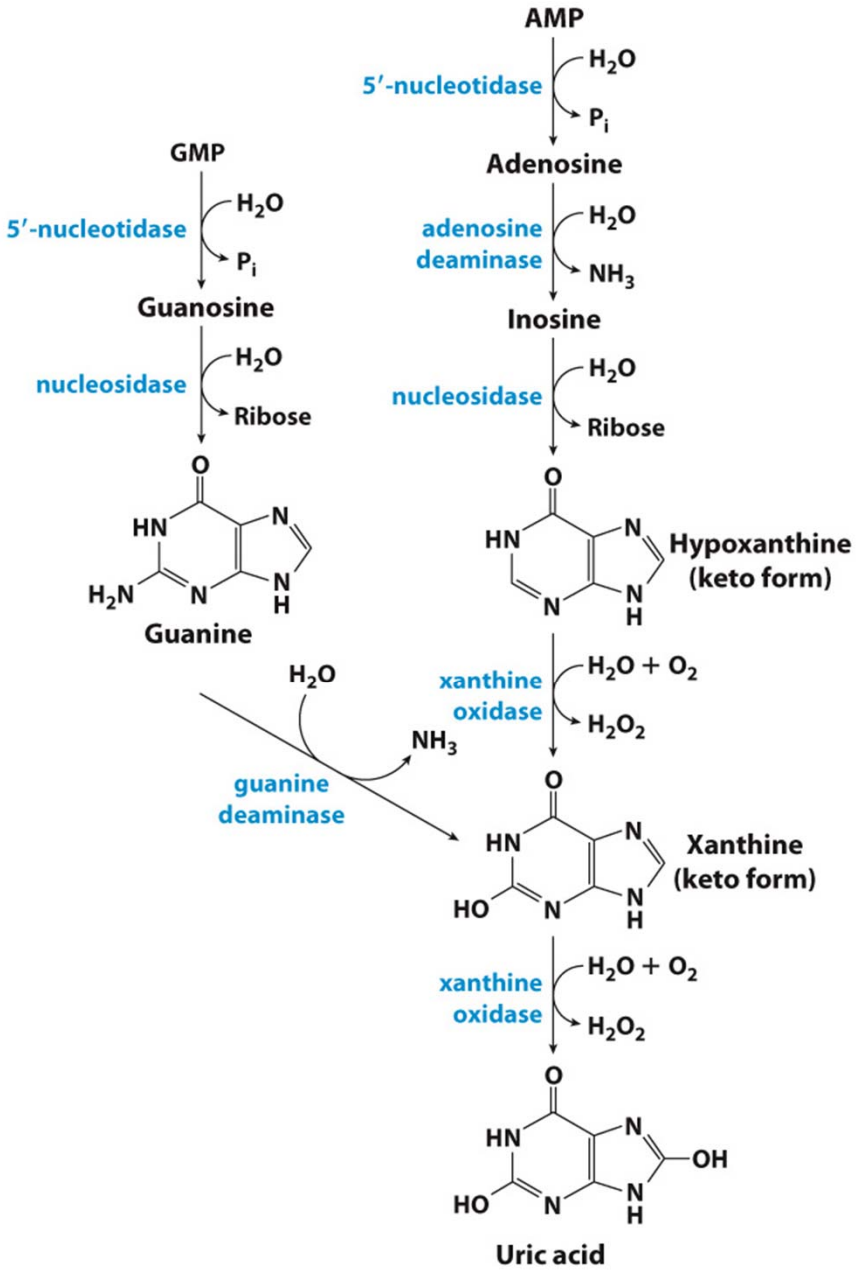
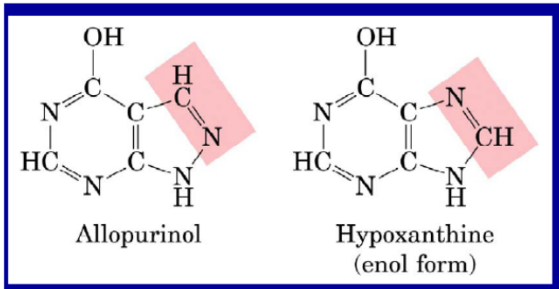


Figure 22-48 part 1
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Excess Uric Acid Seen in Gout

- Painful joints (often in toes) due to deposits of sodium urate crystals
- Primarily affects males
- May involve genetic under-excretion of urate and/or may involve overconsumption of fructose
- Treated with avoidance of purine-rich foods (seafood, liver) or avoidance of fructose
- Also treated with xanthine oxidase inhibitor **allopurinol**





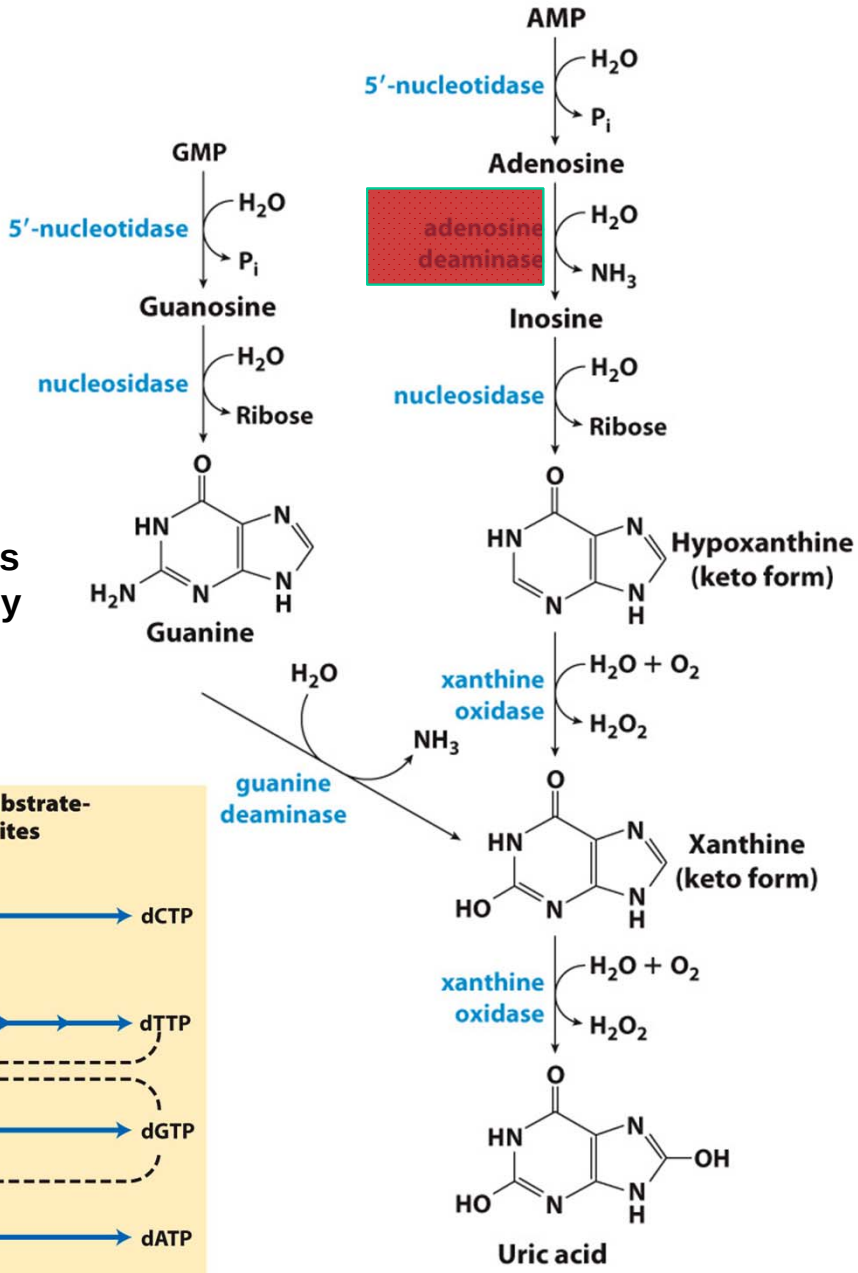
Cristales de ácido úrico en orina (160X)

ADA deficiency causes the severe combined immunodeficiency (SCID)



Adenosine deaminase deficit causes SCID

ADA deficit promotes accumulation of adenosine and as consequence of it, accumulation of dATP which strongly inhibitis ribonucleotide reductase



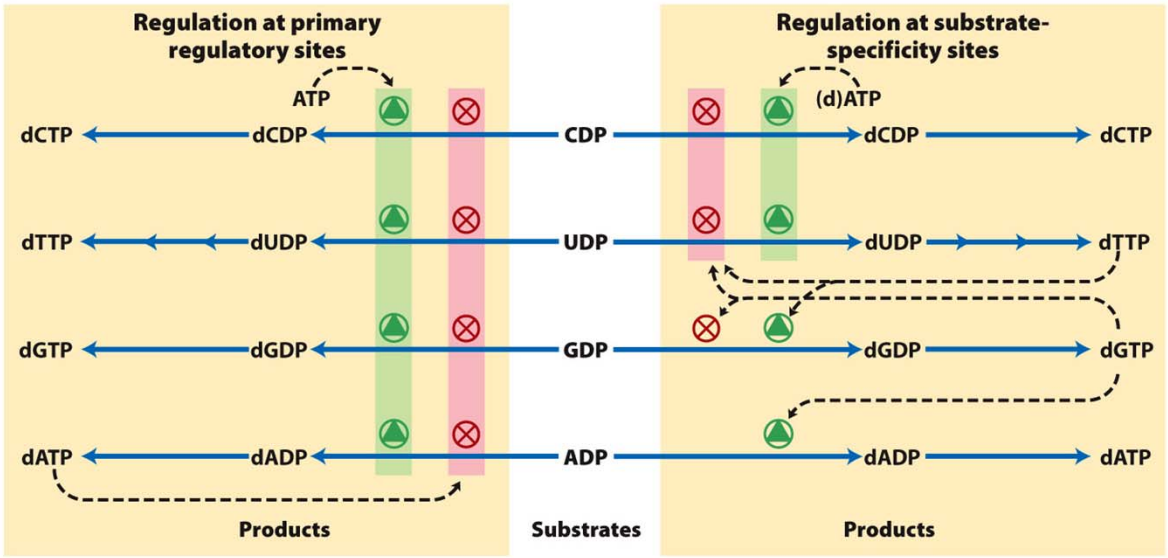


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Catabolism of Pyrimidines

- Leads to NH_4^+ and urea